TCSS 321 – Discrete Structures

Logical Quantifiers

Predicate Logic

- In predicate logic we split a proposition into a subject and a predicate.
  - Until now we have treated propositions as atoms, unbreakable.

Example: Let p be the proposition "Snow is white.".

- In predicate logic we can define a to be the subject "snow" and $W(x)$ to be the predicate "x is white." Then $p = W(a)$ is true.
- We can apply this predicate to other subjects like b = "grass". In this case $W(b)$ is false.
- We can even apply the predicate to subjects that it shouldn't apply to like c = "5". In these cases we define the proposition $W(c)$ to be false.

Universal Quantifier

- A quantifier can be applied to a predicate to create a proposition without a subject.
  - Instead of a subject the quantifier says how many subjects for which the predicate is true.

Definition: The universal quantifier, denoted $\forall_x$, when applied to a predicate $P(x)$, means $P(x)$ for all subjects x.

- Many of the propositions we have considered thus far have had implicit universal quantifiers.
  - When a variable is used in a proposition it is typically assumed we mean for all possible values of the variable. In these cases we can omit the universal quantifier.

Example: Let $O(x)$ be the predicate "is an odd integer".

- The proposition "n is odd if and only if n^2 is odd" can be expressed $\forall_{n\in\mathbb{Z}}\big(O(n) \leftrightarrow O(n^2)\big)$.
- The closure of odd numbers under multiplication can be expressed $\forall_{n\in\mathbb{Z}}\forall_{m\in\mathbb{Z}}\Big(\big(O(n) \wedge O(m)\big) \rightarrow O(nm)\Big)$.

- With the universal quantifier we can express propositions like p = "Every bachelor is unmarried."
  - Let $B(x)$ express the predicate "is a bachelor" and let $M(x)$ be the proposition "is married". Then $p = \forall_x\big(B(x) \rightarrow \neg M(x)\big)$.

Existential Quantifier

Definition: The existential quantifier, denoted $\exists_x$, when applied to a predicate $P(x)$, means $P(x)$ for some subject x.

Example: Let $O(x)$ be the predicate "is an odd integer" and let $E(x)$ be the predicate "is an even integer".

- The proposition "there is an integer that is both even and odd" can be expressed $\exists_{n\in\mathbb{Z}}\big(E(n) \wedge O(n)\big)$. This is a false proposition.
- What does $\exists_{n\in\mathbb{Z}}(n+4=0)$ mean? Is this proposition true?

- Can we express the proposition "there is no largest integer"? Hint: What does it mean to be the largest integer?

$$\neg\big(\exists_{n\in\mathbb{Z}}\forall_{m\in\mathbb{Z}}(n \geq m)\big)$$

Negating Quantifiers

- What are $\neg\forall_x P(x)$ and $\neg\exists_x P(x)$ equivalent to?

- The following are logical equivalences:

$$\neg\forall_x P(x) \equiv \exists_x \neg P(x)$$
$$\neg\exists_x P(x) \equiv \forall_x \neg P(x)$$

- Let's apply these equivalences to our proposition "there is no largest integer".

$$\begin{aligned}&\neg\big(\exists_{n\in\mathbb{Z}}\forall_{m\in\mathbb{Z}}(n \geq m)\big)\\ &\equiv \forall_{n\in\mathbb{Z}}\ \neg\big(\forall_{m\in\mathbb{Z}}(n \geq m)\big)\\ &\equiv \forall_{n\in\mathbb{Z}}\ \exists_{m\in\mathbb{Z}}\neg(n \geq m)\\ &\equiv \forall_{n\in\mathbb{Z}}\ \exists_{m\in\mathbb{Z}}(n < m)\end{aligned}$$

- This proposition states "for every integer there exists an integer that is greater" which is logically equivalent to "there is no largest integer".

Nested Quantifiers

- When the nested quantifiers are of the same type the order of nesting does not matter.

$$\forall_x \forall_y P(x,y) \equiv \forall_y \forall_x P(x,y) \equiv \forall_{x,y} P(x,y)$$
$$\exists_x \exists_y P(x,y) \equiv \exists_y \exists_x P(x,y) \equiv \exists_{x,y} P(x,y)$$

- However, when the nested quantifiers are of different types the order of nesting does matter.

$$\exists_x \forall_y P(x,y) \not\equiv \forall_y \exists_x P(x,y)$$
$$\forall_x \exists_y P(x,y) \not\equiv \exists_y \forall_x P(x,y)$$

Example: What does $\exists_{n\in\mathbb{Z}} \forall_{m\in\mathbb{Z}} (n < m)$ mean?

TCSS 321 – Discrete Structures

Asymptotic Notation

Asymptotic Notation

Definition: Asymptotic notation, also known as “big-Oh” notation, is a notation used to group similar runtime functions together.

- Let $f(n)$ and $g(n)$ be runtime functions.

Definition: A runtime function $f(n) \in O\big(g(n)\big)$, if and only if, there exists positive constants b and n_0 such that for all $n > n_0$ it is the case that $f(n) \leq b \cdot g(n)$.

- Or in predicate logic:

$$f(n) \in O\big(g(n)\big) \leftrightarrow \exists_{b>0} \exists_{n_0>0} \forall_{n>n_0} : f(n) \leq b \cdot g(n)$$

- Or simpler:

$$f(n) \in O\big(g(n)\big) \leftrightarrow \exists_{b,n_0>0} \forall_{n>n_0} : f(n) \leq b \cdot g(n)$$

$O(n)$

- Recall that we analyzed the runtime of Minimum, Search, and String Equals.

$$f(n) = 4n + 1$$
$$g(n) = 3n + 4$$
$$h(n) = 3n + 5$$

- Can we show using the definition that these functions are in $O(n)$?

$O(n)$

Theorem: $f(n) = 4n + 1 \in O(n)$

Proof: We need to find positive constants b and n_0 such that for all $n > n_0$ it is the case that $4n + 1 \leq bn$.

Notice that $4n + 1 \leq 4n + n$ whenever $n \geq 1$. So $4n + 1 \leq 5n$ for all $n > 0$. ■

Theorem: $f(n) = 3n + 4 \in O(n)$

Proof: We need to find positive constants b and n_0 such that for all $n > n_0$ it is the case that $3n + 4 \leq bn$.

Notice that $3n + 4 \leq 3n + n$ whenever $n \geq 4$. So $3n + 4 \leq 5n$ for all $n > 3$. ■

Theorem: $f(n) = 3n + 5 \in O(n)$

Proof: We need to find positive constants b and n_0 such that for all $n > n_0$ it is the case that $3n + 5 \leq bn$.

Notice that $3n + 5 \leq 3n + 5n$ whenever $n \geq 1$. So $3n + 5 \leq 8n$ for all $n > 0$. ■

$O(n^2)$

- Consider some potential runtime functions.

$$f(n) = 7n^2 - 2n + 4$$
$$g(n) = n^2 + 4n - 5$$
$$h(n) = 3n + 5$$

- Can we show using the definition that these functions are in $O(n^2)$?

$O(n^2)$

Theorem: $f(n) = 7n^2 - 2n + 4 \in O(n^2)$

Proof: We need to find positive constants b and n_0 such that for all $n > n_0$ it is the case that $7n^2 - 2n + 4 \leq bn^2$.

Notice that $7n^2 - 2n + 4 \leq 7n^2 + 4 \leq 7n^2 + 4n^2$ whenever $n \geq 1$. So $7n^2 - 2n + 4 \leq 11n^2$ for all $n > 0$. ■

Theorem: $f(n) = n^2 + 4n - 5 \in O(n^2)$

Proof: We need to find positive constants b and n_0 such that for all $n > n_0$ it is the case that $n^2 + 4n - 5 \leq bn^2$.

Notice that $n^2 + 4n - 5 \leq n^2 + 4n \leq n^2 + 4n^2$ whenever $n \geq 1$. So $n^2 + 4n - 5 \leq 5n^2$ for all $n > 3$. ■

Theorem: $f(n) = 3n + 5 \in O(n^2)$

Proof: We need to find positive constants b and n_0 such that for all $n > n_0$ it is the case that $3n + 5 \leq bn^2$.

Notice that $3n + 5 \leq 3n^2 + 5n^2$ whenever $n \geq 1$. So $3n + 5 \leq 8n^2$ for all $n > 0$. ■

Big Theta

Definition: A runtime function $f(n) \in O\big(g(n)\big)$, if and only if, there exists positive constants b and n_0 such that for all $n > n_0$ it is the case that $f(n) \leq b \cdot g(n)$.

Definition: A runtime function $f(n) \in \Omega\big(g(n)\big)$, if and only if, there exists positive constants a and n_0 such that for all $n > n_0$ it is the case that $f(n) \geq a \cdot g(n)$.

- If we put the big-Oh and big-Omega definitions together we get the definition of big-Theta. A big-Theta bound is both an upper and lower bound and is usually called a "tight bound".

Definition: A runtime function $f(n) \in \Theta\big(g(n)\big)$, if and only if, there exists positive constants a, b and n_0 such that for all $n > n_0$ it is the case that $a \cdot g(n) \leq f(n) \leq b \cdot g(n)$.

$\Omega(n)$

- Recall that we analyzed the runtime of Minimum, Search, and String Equals.

$$f(n) = 4n + 1$$
$$g(n) = 3n + 4$$
$$h(n) = 3n + 5$$

- Can we show using the definition that these functions are in $\Omega(n)$?

$\Omega(n)$

Theorem: $f(n) = 4n + 1 \in \Omega(n)$

Proof: We need to find positive constants a and n_0 such that for all $n > n_0$ it is the case that $4n + 1 \geq an$.

Notice that $4n + 1 \geq 4n$ whenever $n \geq 0$. So $4n + 1 \geq 4n$ for all $n > 0$. ■

Theorem: $f(n) = 3n + 4 \in \Omega(n)$

Proof: We need to find positive constants a and n_0 such that for all $n > n_0$ it is the case that $3n + 4 \geq an$.

Notice that $3n + 4 \geq n$ whenever $n \geq 0$. So $3n + 4 \geq n$ for all $n > 0$. ■

Theorem: $f(n) = 3n + 5 \in \Omega(n)$

Proof: We need to find positive constants a and n_0 such that for all $n > n_0$ it is the case that $3n + 5 \geq an$.

Notice that $3n + 5 \geq n$ whenever $n \geq 0$. So $3n + 5 \geq n$ for all $n > 0$. ■

$\Omega(n^2)$

- Consider some potential runtime functions.

$$f(n) = 7n^2 - 2n + 4$$
$$g(n) = n^2 + 4n - 5$$
$$h(n) = 3n + 5$$

- Can we show using the definition that these functions are in $\Omega(n^2)$?
  - Notice $h(n) \notin \Omega(n^2)$! So we cannot prove this one.

$\Omega(n^2)$

Theorem: $f(n) = 7n^2 - 2n + 4 \in \Omega(n^2)$

Proof: We need to find positive constants a and n_0 such that for all $n > n_0$ it is the case that $7n^2 - 2n + 4 \geq an^2$.

Notice that $7n^2 - 2n + 4 \geq 7n^2 - 2n \geq 7n^2 - 2n^2$ whenever $n \geq 1$. So $7n^2 - 2n + 4 \geq 5n^2$ for all $n > 0$. ■

Theorem: $f(n) = n^2 + 4n - 5 \in \Omega(n^2)$

Proof: We need to find positive constants a and n_0 such that for all $n > n_0$ it is the case that $n^2 + 4n - 5 \geq an^2$.

Notice that $n^2 + 4n - 5 \geq n^2 - 5 \geq \frac{4}{5}n^2$ whenever $n \geq 5$. So $n^2 + 4n - 5 \geq \frac{4}{5}n^2$ for all $n > 4$. ■

$\Theta(n)$ and $\Theta(n^2)$

- In conclusion we have shown the following tight bounds:

$$4n + 1 \in \Theta(n)$$
$$3n + 4 \in \Theta(n)$$
$$3n + 5 \in \Theta(n)$$

$$7n^2 - 2n + 4 \in \Theta(n^2)$$
$$n^2 + 4n - 5 \in \Theta(n^2)$$

TCSS 321 – Discrete Structures

The Limit Test

Little-Oh Notation

Definition: A runtime function $f(n) \in o(g(n))$, if and only if, for every positive constant b there exists a positive constant n_0 such that for all $n > n_0$ it is the case that $f(n) < bg(n)$.

Definition: A runtime function $f(n) \in \omega(g(n))$, if and only if, for every positive constant a there exists a positive constant n_0 such that for all $n > n_0$ it is the case that $f(n) > ag(n)$.

- As a rule of thumb we can remember these relationships.
  - $f(n) \in O(g(n)$ means $f(n) \leq g(n)$
  - $f(n) \in o(g(n)$ means $f(n) < g(n)$
  - $f(n) \in \Theta(g(n)$ means $f(n) = g(n)$
  - $f(n) \in \Omega(g(n)$ means $f(n) \geq g(n)$
  - $f(n) \in \omega(g(n)$ means $f(n) > g(n)$

The Limit Test

Theorem: If $f(n)$ and $g(n)$ are two runtime functions then

1. If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty$ then $f(n) \in \omega\big(g(n)\big)$.
2. If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$ then $f(n) \in o\big(g(n)\big)$.
3. If $\lim_{n\to\infty} \frac{f(n)}{g(n)} = c$ for some non-zero c then $f(n) \in \Theta\big(g(n)\big)$.

- We can use the limit test to quickly determine the relationship between two functions.

The Limit Test

Theorem: $4n^2 - n \in \omega(n)$

Proof: We will use the limit test.

$$\lim_{n\to\infty} \frac{4n^2 - n}{n} = \lim_{n\to\infty} \left(\frac{4n^2}{n} - \frac{n}{n} \right)$$

$$= \lim_{n\to\infty} (4n) - \lim_{n\to\infty} (1)$$

$$= \infty$$

So by the limit test $4n^2 - n \in \omega(n)$. ■

The Limit Test

Theorem: $6n^3 + 5n - 12 \in \Theta(n^3)$

Proof: We will use the limit test.

$$\lim_{n\to\infty} \frac{6n^3 + 5n - 12}{n^3} = \lim_{n\to\infty} \left(\frac{6n^3}{n^3} + \frac{5n}{n^3} - \frac{12}{n^3} \right)$$

$$= \lim_{n\to\infty} \left(6 + \frac{5}{n^2} - \frac{12}{n^3} \right)$$

$$= \lim_{n\to\infty} (6) + \lim_{n\to\infty} \left(\frac{5}{n^2} \right) - \lim_{n\to\infty} \left(\frac{12}{n^3} \right)$$

$$= 6 + 0 - 0 = 6$$

So by the limit test $6n^3 + 5n - 12 \in \Theta(n^3)$. ■

Logarithms dominated by Polynomials

Theorem: $\log_2(n) \in o(n^d)$ for $d > 0$.

Proof: We will use the limit test. Let $d > 0$.

$$\lim_{n\to\infty} \frac{\log_2(n)}{n^d} = \lim_{n\to\infty} \frac{\frac{1}{\ln 2\, n}}{dn^{d-1}}$$

$$= \lim_{n\to\infty} \frac{1}{\ln 2\, n d n^{d-1}}$$

$$= \lim_{n\to\infty} \frac{1}{\ln 2\, d n^{d}}$$

$$= 0$$

So by the limit test $\log_2(n) \in o(n^d)$ for $d > 0$. ■

Polynomials dominated by Exponentials

Theorem: $n \in o\left((1+a)^n\right)$ for $a > 0$.

Proof: We will use the limit test. Let $a > 0$.

$$\lim_{n\to\infty} \frac{n}{(1+a)^n} = \lim_{n\to\infty} \frac{1}{\ln(1+a)\cdot(1+a)^n}$$

$$= 0$$

So by the limit test $n \in o\left((1+a)^n\right)$ for $a > 0$. ■

Tight Bounds

Theorem: If $f(n) \in O\big(g(n)\big)$ and $f(n) \in \Omega\big(g(n)\big)$ then $f(n) \in \Theta\big(g(n)\big)$.

- We could prove this with the definitions or take this as an axiom. You will prove a theorem like this on the Assignment 5.
- When we have the same upper bound (O) and lower bound (Ω) this establishes a tight bound (Θ).

- Can we find and prove tight bounds for these functions and then place them in order?

$$f_0(n) = n^2 - 6n + 5$$
$$f_1(n) = (2n + 5)^8$$
$$f_2(n) = \sqrt[4]{n^2 + n - 5}$$
$$f_3(n) = \log_2(\log_2(n) + 1)$$
$$f_4(n) = 16^{\log_2 n}$$